

Sai Sandeep Kumar Vijayarao

AI Engineer | Agents, Retrieval & LLM Systems | MS CS, Northeastern (May 2027)

San Jose, CA | +1 (408) 210-6329 | sandeepvijayarao09@gmail.com | linkedin.com/in/sandeepvijayarao |
github.com/sandeepvijayarao09 | Portfolio

AI engineer building production systems on foundation models: agent loops with real tools, RAG over a vector store, evaluation harnesses with stated case counts, and a router that keeps most traffic off frontier models. Shipped a multi-model platform load-tested to 10K+ requests/day.

EXPERIENCE

Northeastern University

Graduate Teaching Assistant, Computer Vision, GenAI & HCI

San Jose, CA

Sep 2025 - Present

- Built the evaluation harnesses and lab infrastructure behind three graduate courses running agentic AI, on-device inference and multimodal workloads, carrying **100+ students to benchmarked on-deadline submissions** across two semesters.

Trominosoft

AI Developer

Hyderabad, India

Feb 2025 - Aug 2025

- Engineered the orchestration, retrieval and tool-use layers of a multi-model AI platform in Python on Docker and Kubernetes on GCP, **load-tested to 10K+ requests/day** behind zero-downtime update workflows.
- Instrumented end-to-end observability across that platform's request paths, then profiled and optimized model serving against what it exposed, **cutting production request latency 30%**.
- Built DPO and LoRA fine-tuning pipelines against models already serving production traffic, **reducing hallucination rate 38%** without taking the platform down.

Infosys Springboard

Artificial Intelligence Intern

India

Sep 2024 - Dec 2024

- Designed the async API bridges unifying three separate services, BLIP for vision, GPT-3.5 for language and Google TTS for speech, behind one platform, **cutting task completion time 50%**.
- Benchmarked INT4/INT8 quantization and LoRA fine-tuning on 7B-parameter models via vLLM, **achieving 3.2x inference throughput** at under 2% quality degradation.

KEY PROJECTS

Dual LLM Routing System: Cost-Aware Model Router | Python, scikit-learn, SQLite [link]

- Designed a 3-layer router that dispatches between a lightweight local model and a frontier fallback, **reaching 92.6% accuracy on a 1000-case eval** at a 14.7% cascade rate, so most traffic costs nothing in frontier tokens.

Anu: On-Device Agentic Assistant | Apple Swift Student Challenge 2026 Winner

Swift, CoreML, LiteRT [link]

- Shipped an iOS agent running Gemma 4 E4B fully on-device via LiteRT; a TaskClassifier **scoring 86.7% on a 60-case eval** routes intents through a ReAct loop over calculator, web-search and cloud-escalation tools, behind a two-layer PII scrub.

Agentic Second Brain: Gmail Extraction Agent | Python, FastAPI, Gemini, Next.js [link]

- Built a Gemini extraction agent that parses unstructured order confirmations into a structured SQLite database, **queried in natural language** through a FastAPI and Next.js chat UI on DigitalOcean.

Agent-Certifier: Agent Testing & Certification Service | Python, FastAPI, Next.js [link]

- Built a FastAPI and Next.js service that statically analyses agent codebases it did not author, running **60 tests across 6 categories** and returning a weighted certification level.

Northeastern University | MS, Computer Science

San Jose, CA | Sep 2025 - May 2027

R.M.K. Engineering College | B.E. Computer Science & Engineering

Chennai, India | Oct 2021 - May 2025

TECHNICAL SKILLS

Languages: Python, TypeScript, Swift, SQL, Bash

LLM & Agents: OpenAI/Anthropic/Gemini APIs, LangChain, Google ADK, MCP Protocol, ReAct, multi-agent pipelines, tool use, agent memory, prompt engineering

Retrieval: RAG, ChromaDB, vector search, embedding retrieval, sentence-transformers, SQLite

Serving & Inference: vLLM, INT4/INT8 quantization, on-device inference, LiteRT, CoreML, KV-cache

Backend & Infrastructure: FastAPI, Node.js, Next.js, REST APIs, Docker, Kubernetes, GCP, DigitalOcean, CI/CD, Git, pytest

Evaluation & Safety: ROUGE/BLEU/BERTScore, static analysis, PII scrubbing, OAuth2, AES-256-GCM

EDUCATION